

Maincraft Technology

AIML Internship Task 4: Classification Models, Evaluation Metrics & Handling Imbalanced Data

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Internship Domain: Artificial Intelligence & Machine Learning

Introduction

The objective of this task was to build and evaluate classification models using the Breast Cancer Wisconsin Dataset. The project focused on evaluating different classification algorithms using multiple performance metrics and selecting the most suitable model. Techniques for handling class imbalance were also applied to improve reliability.

The dataset contains numerical features extracted from breast cell images. The target variable classifies tumors as Malignant or Benign. The data was divided into training and testing sets, and multiple machine learning models were trained and evaluated.

Metric Selection

Accuracy, Precision, Recall, and F1-Score were used to evaluate the classification models. Accuracy measures the overall percentage of correct predictions. Precision measures the correctness of positive predictions, while Recall measures the ability to identify actual positive cases. F1-Score provides a balanced evaluation by combining Precision and Recall.

A Confusion Matrix was generated to visualize prediction results. ROC Curve and AUC Score were also used to measure the model's ability to distinguish between classes.

Imbalance Handling

To reduce bias caused by class imbalance, Logistic Regression was trained using `class_weight='balanced'`. This assigns higher importance to minority class samples during training and improves classification performance for underrepresented cases. The balanced model was compared with the standard Logistic Regression model and the Decision Tree Classifier.

Final Model Decision

Three models were evaluated: Logistic Regression, Balanced Logistic Regression, and Decision Tree Classifier. Their performances were compared using Accuracy, Precision, Recall, F1-Score, and ROC-AUC.

The model with the best overall evaluation metrics and strongest generalization ability was selected as the final model. This comparison demonstrated the importance of evaluating multiple models before selecting the final solution.

Conclusion

This task provided practical experience in classification model evaluation, performance comparison, handling imbalanced data, and model selection. Using multiple evaluation metrics resulted in a more reliable assessment than using accuracy alone, leading to a robust and effective machine learning model.